

# ***Lab Rotation Report***

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**Abstract:**

This project investigates the impact of deafness and lateralization on the structural connectivity of the auditory pathway. Diffusion tensor imaging (DTI) was used to analyze white matter tracts connecting the inferior colliculus to the primary auditory cortex in deaf individuals. Diffusion metrics were compared within the same group for left and right hemispheres, as well as across groups for the same hemisphere. Two primary groups were examined: individuals with left or right deafness and individuals categorized as left-lateralized (with dominant speech processing in the left hemisphere) or bi-lateralized (with speech processing distributed across both hemispheres). It was hypothesized that deafness would alter hemispheric asymmetry in these pathways, indicating neural reorganization. The findings contribute to a deeper understanding of neuroplasticity in the auditory system in the absence of auditory input.

**Background:**

Diffusion tensor imaging (DTI) is an advanced MRI technique used to study white matter structure in the brain. It estimates the trajectory of white matter pathways by measuring water molecule diffusion within tissue. The directionality of water diffusion, known as anisotropy, is quantified using fractional anisotropy (FA), while mean diffusivity (MD) represents the overall magnitude of diffusion. Additional diffusion metrics include axial diffusivity (AD), which reflects diffusion along the primary fiber direction, and radial diffusivity (RD), which indicates diffusion perpendicular to the fiber direction (Profant, 2014).

Tractography, a computational method, reconstructs white matter pathways from DTI data, allowing for visualization of connectivity between brain regions. The auditory pathway is a network of structures responsible for processing sound (Profant, 2014). Studies using MRI and DTI have examined the auditory system in various populations, including individuals with hearing loss. Research suggests that aging and hearing loss alter white matter pathways and gray matter structures in the auditory cortex (Profant, 2014).

Lateralization refers to the specialization of brain functions in one hemisphere. Auditory processing, particularly for speech, is usually left-lateralized (Wendt, 2021). However, congenital deafness may modify this asymmetry due to compensatory neural changes (Profant, 2014). Some studies suggest that gray matter asymmetry remains unchanged in deaf individuals (Tremblay & Dick, 2016), while others report changes in white matter integrity and connectivity (Profant, 2014). Investigating the diffusion properties of white matter tracts between the brainstem and auditory cortex can help understand how deafness impacts brain structure and speech processing networks.

## **Aims of Study**

This study aims to examine the effects of deafness and lateralization on white matter connectivity in the auditory pathway using diffusion tensor imaging (DTI). The main objectives are:

1. To analyze the white matter tracts linking the inferior colliculus to the primary auditory cortex.

2. To compare diffusion metrics (FA, MD, AD, RD) between the left and right hemispheres within the same group.
3. To compare diffusion metrics within the same hemisphere between different groups (left-deaf vs. right-deaf left-lateralized with dominant speech processing vs. bi-lateralized individuals with speech processing distributed across hemispheres).

### **Hypotheses:**

1. Deafness leads to structural changes in the auditory pathway, altering hemispheric asymmetry in white matter connectivity.
2. Differences in diffusion metrics, including FA, MD, AD, and RD, are expected between the left and right hemispheres within each group.
3. Different lateralization patterns, particularly in speech processing, influence diffusion metric values in the auditory pathway.

### **Methods:**

**Participants:** 27 individuals with unilateral deafness participated in this study. Participants were categorized into two main groups:

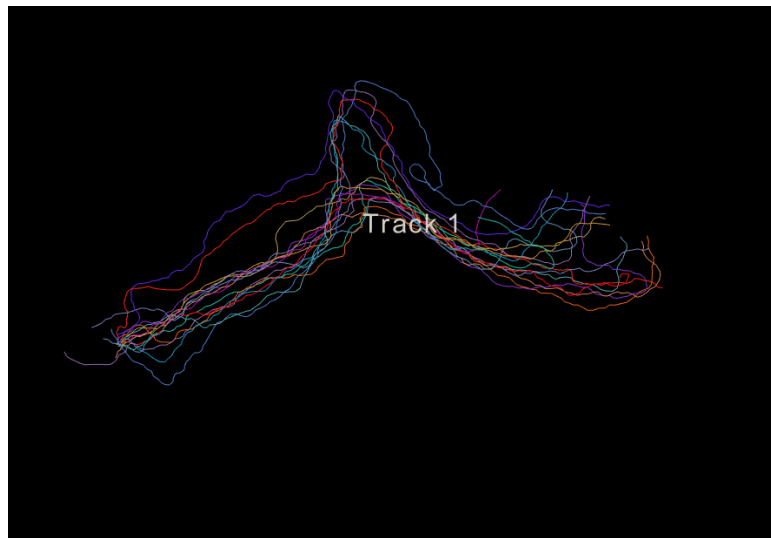
- **Deafness group:** Left-deaf and right-deaf individuals.
- **Lateralization group:** Individuals classified as left-lateralized (with left-hemisphere dominant speech processing) or bi-lateralized (with speech processing distributed across both hemispheres) based on laterality index calculations.

**Tractography:** Probabilistic tractography was performed using DIPY.

- **Seed Generation:** Seed points were generated for tractography in both the inferior colliculus (IC) and the primary auditory cortex.
- **Target Regions:** Both inferior colliculus and primary auditory cortex were defined as the target region
- **Pathway Tracking:** Deterministic or probabilistic tractography algorithms were used to reconstruct the white matter tracts connecting the inferior colliculus to the primary auditory cortex in each hemisphere.
  1. **Stopping Criteria:** A stopping criterion was computed for tractography in the left and right hemispheres based on Generalized Fractional Anisotropy (GFA) from the CsaOdfModel.
  2. **Probabilistic Direction Getter:** A probabilistic direction getter was computed for the left and right hemispheres using Constrained Spherical Deconvolution (CSD).
  3. **Streamline Generation:** Streamlines were generated from IC seeds and AC seeds.
  4. **Streamline Filtering:** Streamlines passing through a specific target mask (frontal lobe mask) were excluded. IC streamlines that reached AC and AC streamlines that reached IC were filtered.
  5. **Streamline Combination:** IC to AC and AC to IC streamlines were combined.
  6. **Streamline Retention:** Streamlines with high confidence were retained based on clustering and length filtering.



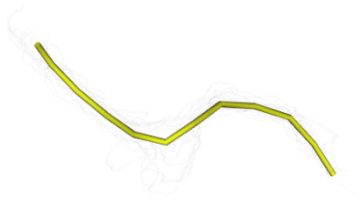
*Figure 1: Right tract from inferior colliculus to primary auditory cortex*



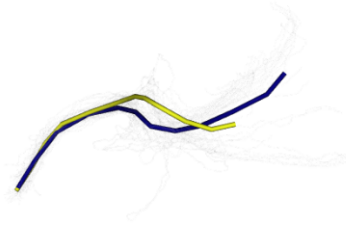
*Figure 2: Left tracts from inferior colliculus to primary auditory cortex*

**Bundle Creation:** Streamlines clustered into bundles using QuickBundles algorithm.

1. **Streamline Resampling:** Streamlines were resampled to ensure they had the same number of points.
2. **Metric Definition:** A metric for clustering was defined as the average Euclidean distance between points.
3. **QuickBundles Clustering:** QuickBundles clustering was applied with a threshold of 10mm.

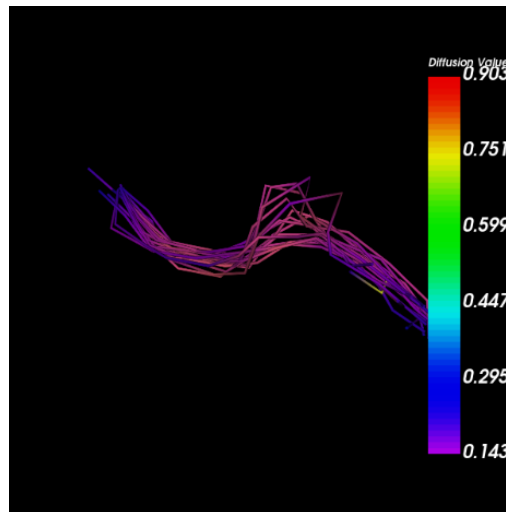


*Figure 3: Left Bundle*



*Figure 4: Right Bundles*

**Diffusion Metrics:** Diffusion metric values (FA, AD, MD, RD) were projected for each streamline in the bundles for the left and right hemispheres. These values were exported to a CSV file for statistical analysis.



*Figure 5: FA values on a bundle*

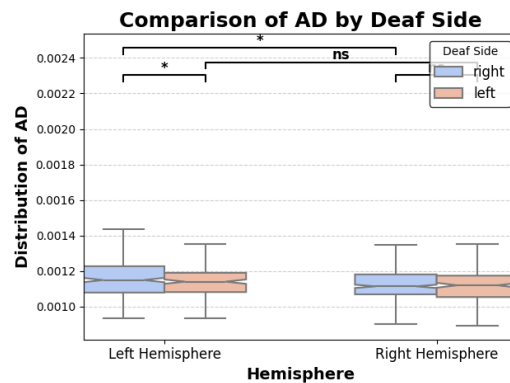


## Statistical analysis

- 1. Laterality Index:** A laterality index was calculated to classify participants as left-lateralized, right-lateralized, or bilateral:
- 2. Auditory Pathway Metrics:** Diffusion metric values (FA, AD, MD, RD) were analyzed for the left and right auditory pathways.
- 3. Independent and Paired t-tests:** Independent and paired t-tests were performed. Paired t-tests to assess differences in diffusion metrics between hemispheres within the same group. Independent t-tests to compare diffusion metrics in the same hemisphere between different groups.

## Results:

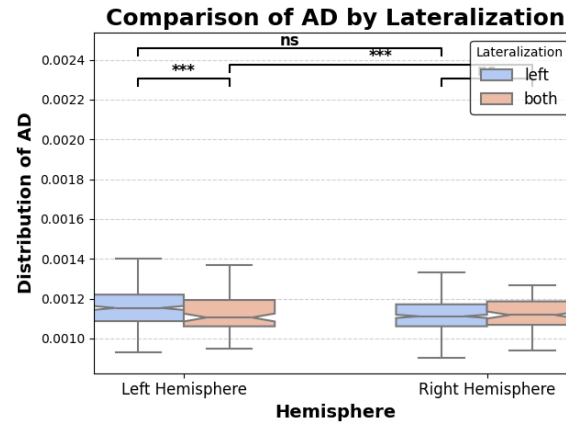
### 1. Axial Diffusivity (AD) Comparisons



*Figure 6: Comparison of AD by Deaf Side*

- A significant difference in AD values was observed between left-deaf and right-deaf individuals in the left hemisphere.

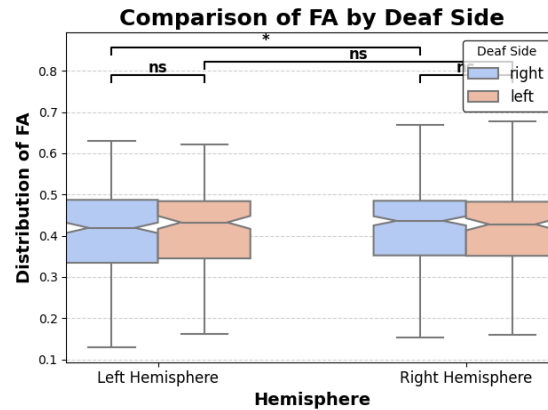
- For right-deaf individuals, there was a significant difference in AD values between the left and right hemispheres.



*Figure 7: Comparison of AD by Lateralization*

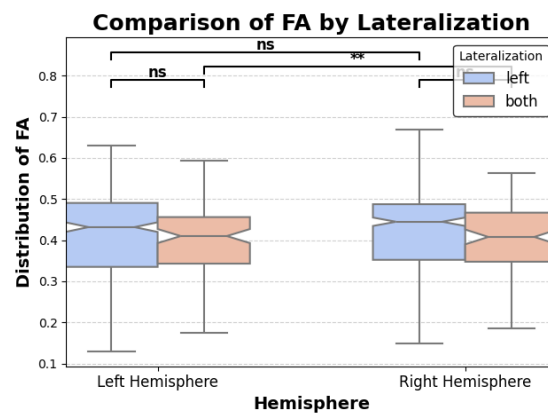
- A significant difference in AD values was found between left-lateralized individuals (with dominant left-hemisphere speech processing) and bi-lateralized individuals (with speech processing distributed across hemispheres) in the left hemisphere.
- A significant difference in AD values was observed between the left and right hemispheres for bi-lateralized individuals (with speech processing distributed across hemispheres).

## 2. Fractional Anisotropy (FA) Comparisons



*Figure 8: Comparison of FA by Deaf Side*

- For right-sided deaf individuals, there was a significant difference in FA values between hemispheres.
- No significant differences were found in the other comparisons.



*Figure 9: Comparison of FA by Lateralization*

- A significant difference in FA values was found between the left and right hemispheres for bi-lateralized individuals, suggesting distinct white matter organization supporting distributed speech processing.
- No significant differences were found in other comparisons.

### 3. Mean Diffusivity (MD) Comparisons

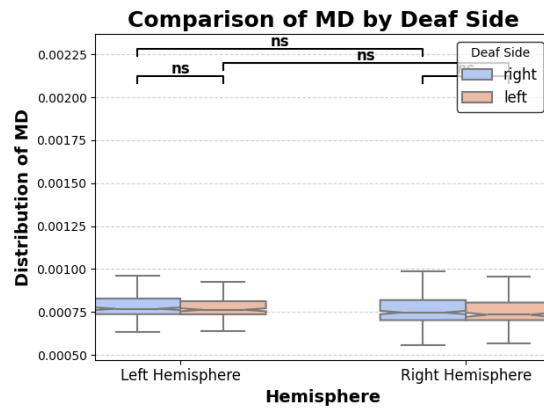


Figure 10: Comparison of MD by Deaf Side

- No significant differences were found in MD values between hemispheres for left- and right-deaf individuals.
- No significant differences were observed within the same hemisphere between left and right deaf individuals.

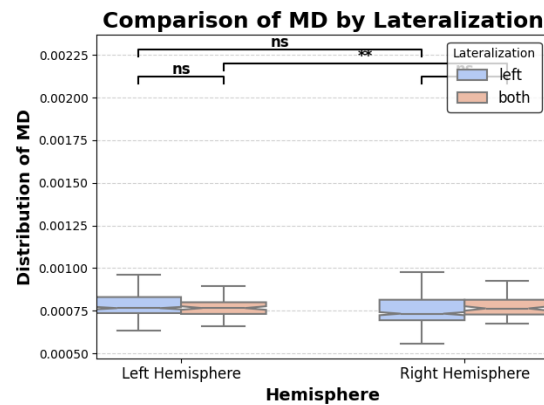
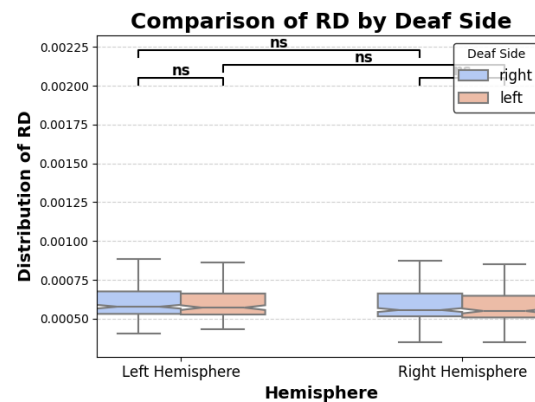


Figure 11: Comparison of MD by Lateralization

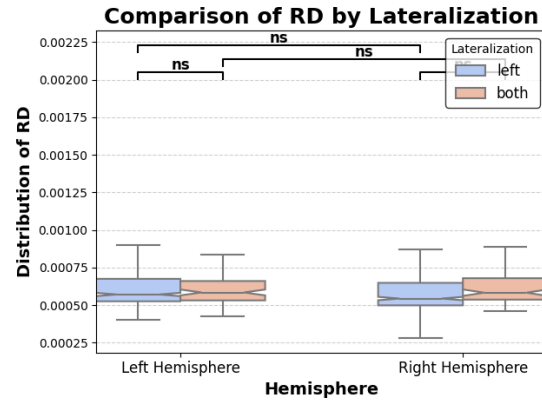
- A significant difference in MD values was observed between the left and right hemispheres for bi-lateralized individuals, indicating structural differences related to speech processing distribution.
- No significant differences were observed for other comparisons.

#### 4. Radial Diffusivity (RD) Comparisons



*Figure 12: Comparison of RD by Deaf Side*

- No significant differences in RD values were found between hemispheres for left- and right-deaf individuals.
- No significant differences were observed within the same hemisphere between left and right deaf individuals.



*Figure 13: Comparison of RD by Lateralization*

- No significant differences in RD values were found between hemispheres for left- and bi-lateralized individuals.
- No significant differences were observed within the same hemisphere between left- and bi-lateralized individuals, suggesting that RD may not be strongly influenced by speech processing lateralization.

### **Discussion:**

The findings suggest that deafness and lateralization affect white matter connectivity in the auditory pathway, particularly in axial diffusivity and fractional anisotropy. These changes indicate potential neuroplastic adaptations in response to auditory deprivation. The significant differences observed in AD and FA support the hypothesis that deafness alters hemispheric asymmetry, and that lateralization influences white matter diffusion properties. However, the lack of significant differences in MD and RD suggests that some aspects of white matter microstructure remain preserved. Future studies should incorporate a control group with normal hearing for comparison.

## References

Profant, O. (2014). Structural and functional changes in the auditory system in hearing loss. *Hearing Research*, 307, 4-12.

Tremblay, K. L., & Dick, F. (2016). Auditory neuroplasticity: The effects of training and sensory deprivation. *Neuropsychologia*, 87, 88-97.

Wendt, M. (2021). Hemispheric specialization in auditory processing: A review of evidence. *Neuroscience & Biobehavioral Reviews*, 125, 1-15.